

Stat 145 (Multivariate Statistics)

Laboratory Exercise No. 4

INSTRUCTIONS: Answer the following as indicated.

The school system of a major city wanted to determine the characteristics of a great teacher, and so they asked 120 students to rate the importance of each of the following criteria using a Likert scale of 1 to 10 with 10 representing that a particular characteristic is extremely important and 1 representing that the characteristic is not important. The criteria are:

- Setting high expectations for the students
- Entertaining
- Able to communicate effectively
- Having expertise in their subject
- Able to motivate
- Caring
- Charismatic
- Having a passion for teaching
- Friendly and easy-going

The data is contained in the Excel file **Stat 145 Lab 4.xlsx** in the GitHub repo.

1. Make sure to clean the data of missing values, if any.
2. Perform PCA using the *prcomp()* function.
 - a. Decide on the number of components to retain. Explain your decision. How much variance is accounted for the retained components?
 - b. Write the equations of the retained principal components. Round the coefficients to two (2) decimal places.
 - c. Generate the biplot of the variables using retained components. Which variables are negatively correlated? positively correlated?